

LEC climatology: published article versus corrected rerun

A figure-by-figure reconstruction using the complete archived legacy population and the validated LorenzCycleToolKit 2.0.0 rerun

Before population	6,789 cyclones / 25,000 lifecycle rows
After population	3,820 cyclones / 15,829 lifecycle rows
Intense clustering	5 groups on PCs 1-8; n=1,744 -> 1,486
Article source commit	3dc622ed0efb
Figure files	20 PNG + 20 vector PDF

Interpretation boundary

This is the literal article-to-rerun comparison. Because the before and after populations differ, visible changes combine the toolkit correction and the change from 6,789 archived systems to 3,820 validated complete-lifecycle reruns. Use the separate paired 3,820-cyclone report when the objective is to isolate the correction alone.

Reproduction audit

The legacy total-lifecycle EOF calculation uses all archived periods for every cyclone, matching the original source workflow. Primary-phase figures use the exact incipient, intensification, mature and decay rows. Corrected EOFs are fitted independently, then reordered and sign-aligned to the legacy loading patterns for visual comparison.

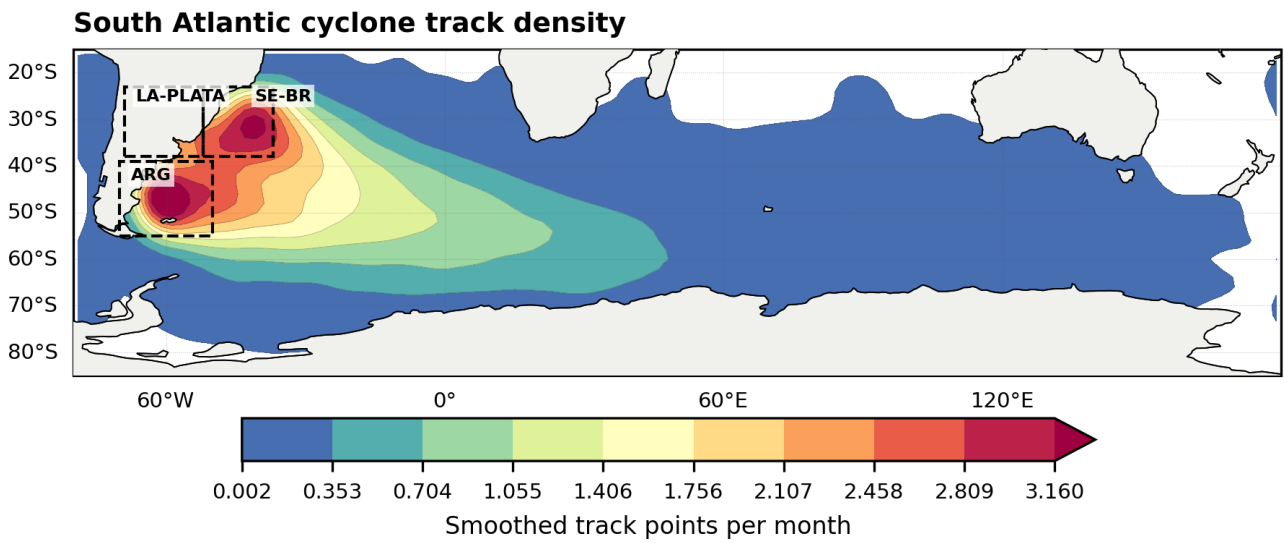
Numerical landmarks

EOF	Before variance	After variance	Matched after rank	Pattern correlation
1	28.304%	27.684%	1	0.798
2	11.018%	12.022%	2	0.961
3	10.925%	10.510%	3	0.579
4	8.188%	7.572%	5	0.767

Intense-cyclone groups

Version	Cluster	n	Mean max. vorticity	Median max. vorticity
before	1	662	9.96	9.65
before	2	332	9.91	9.65
before	3	290	10.70	10.33
before	4	248	10.59	10.41
before	5	212	10.36	10.22
after	1	647	9.84	9.60
after	2	266	10.73	10.57
after	3	229	10.44	10.16
after	4	206	10.62	10.47
after	5	138	10.44	10.21

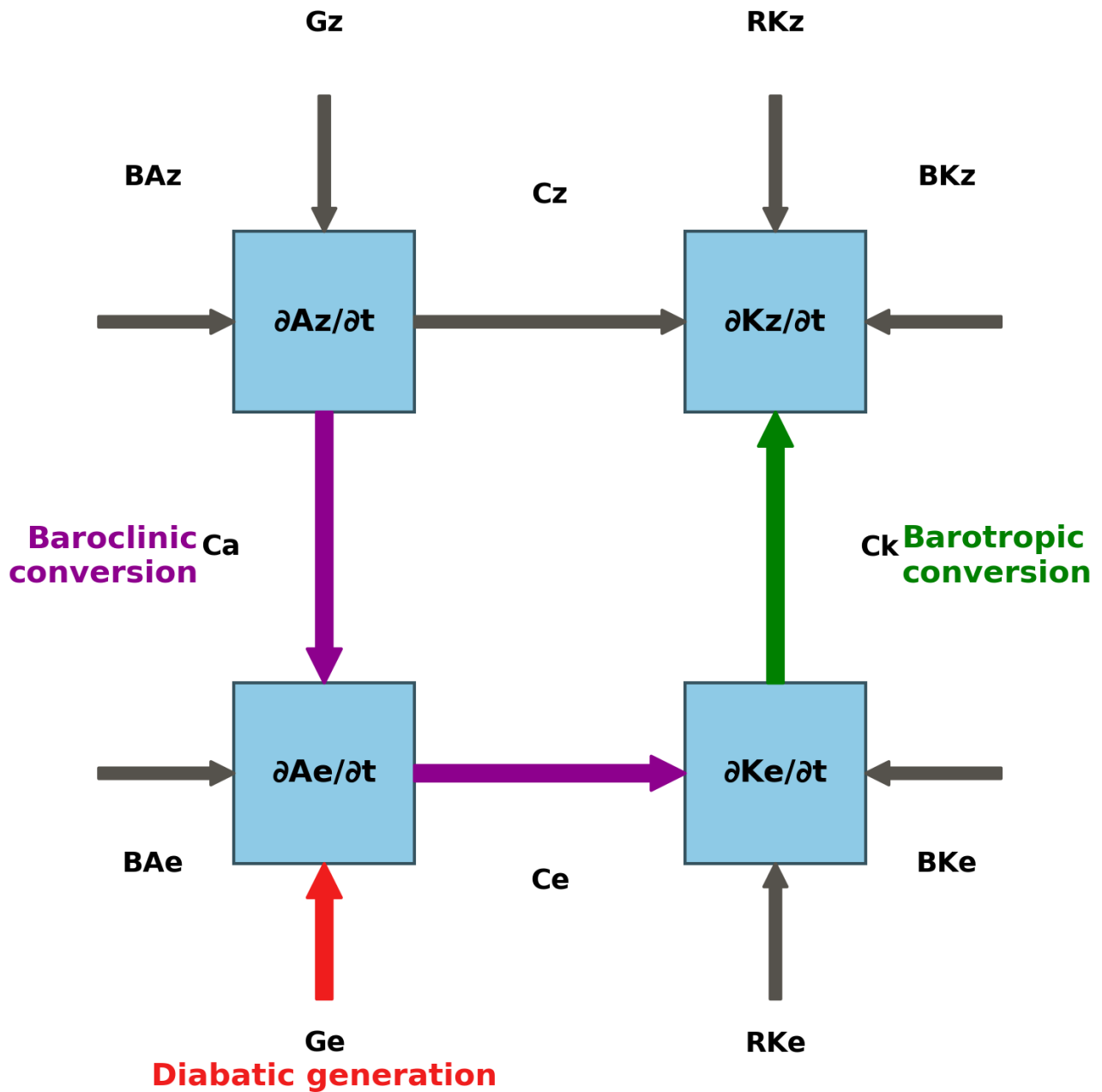
Figure 1



Track density and the three genesis regions; this track-only reference is common to both datasets.

Figure 2

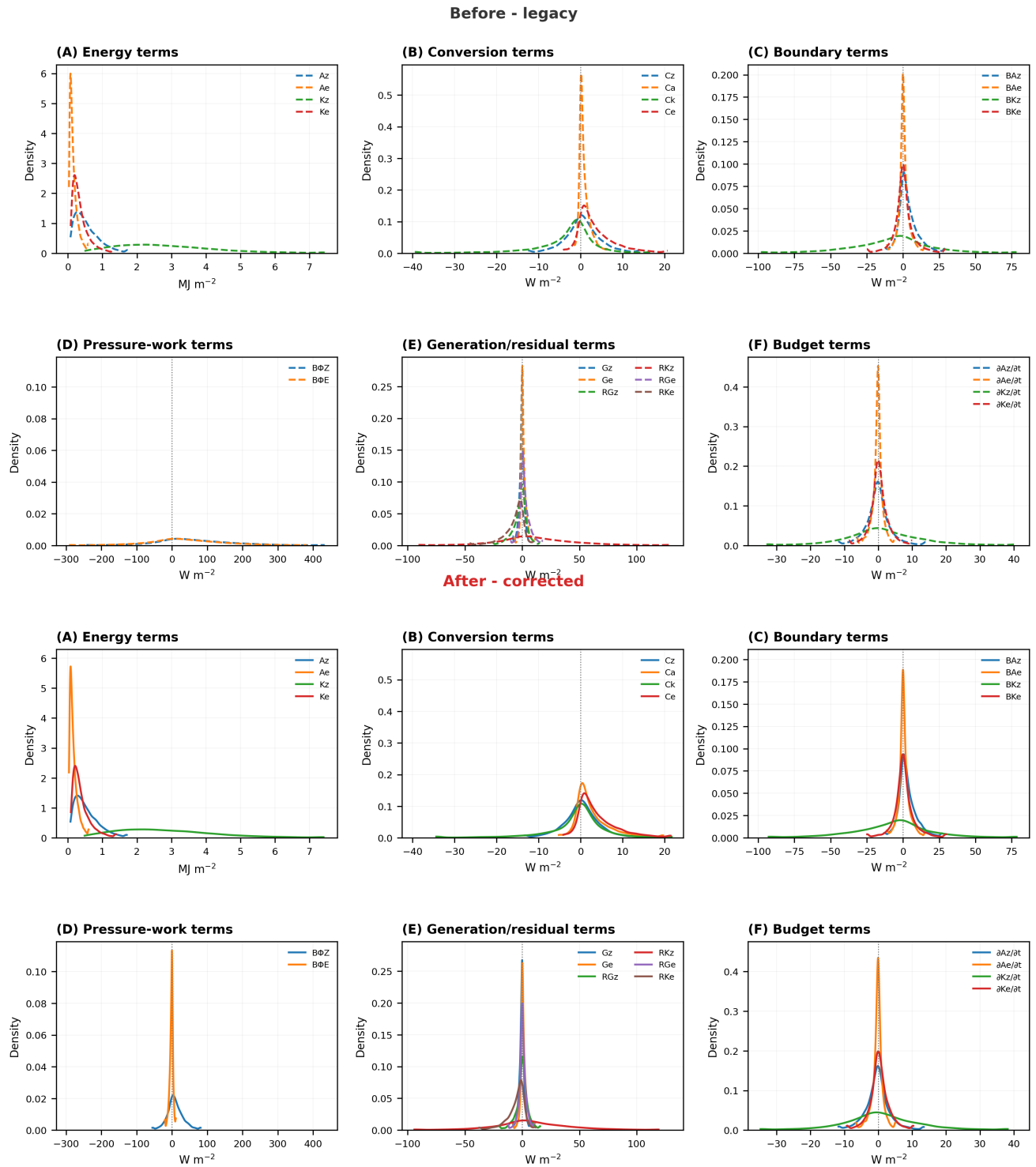
Lorenz Energy Cycle reference diagram



Reference four-box Lorenz Energy Cycle diagram; this conceptual panel is common to both datasets.

Figure 3

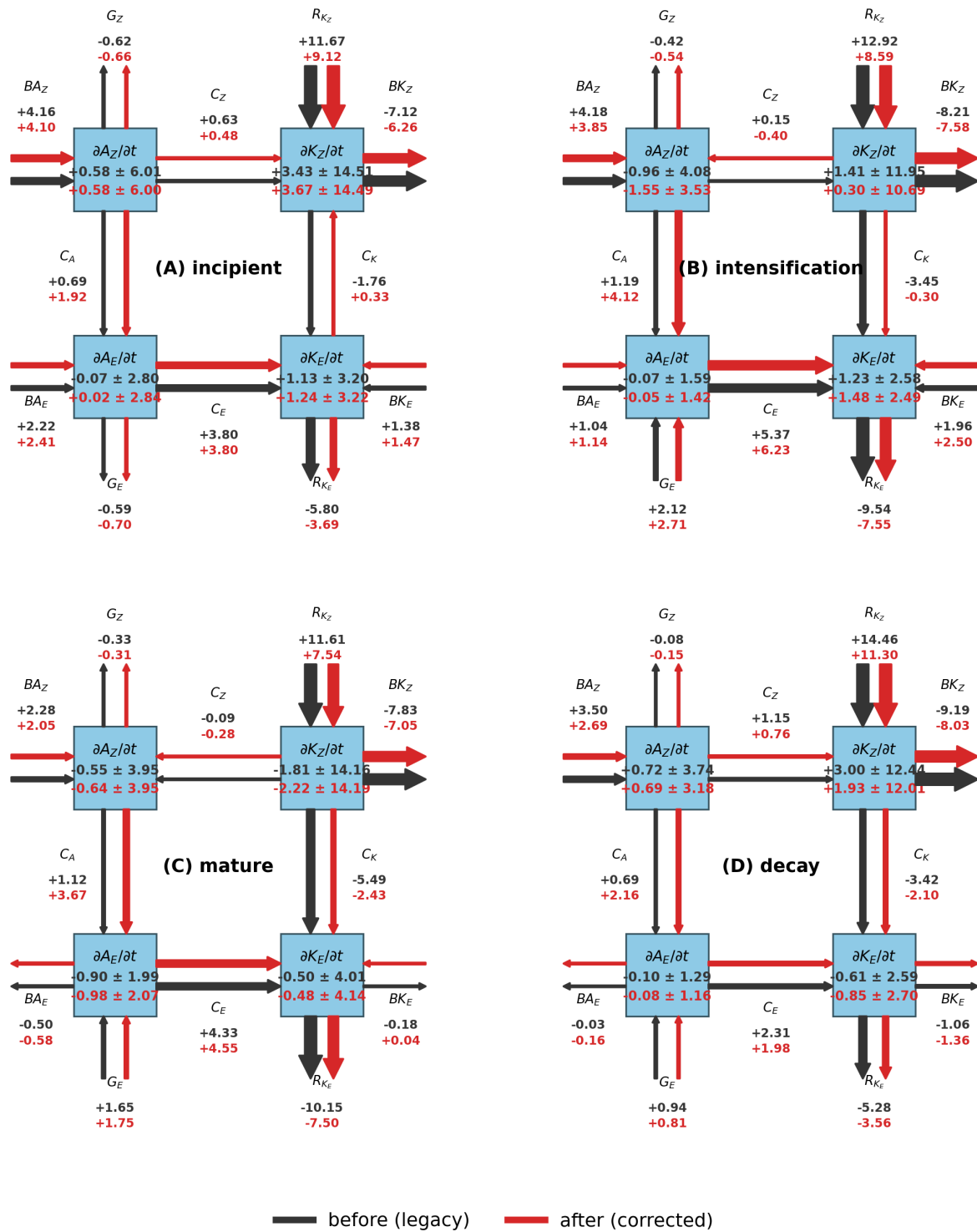
LEC term distributions before and after the toolkit correction



LEC term distributions from the full published legacy archive (upper half) and corrected rerun (lower half).

Figure 4

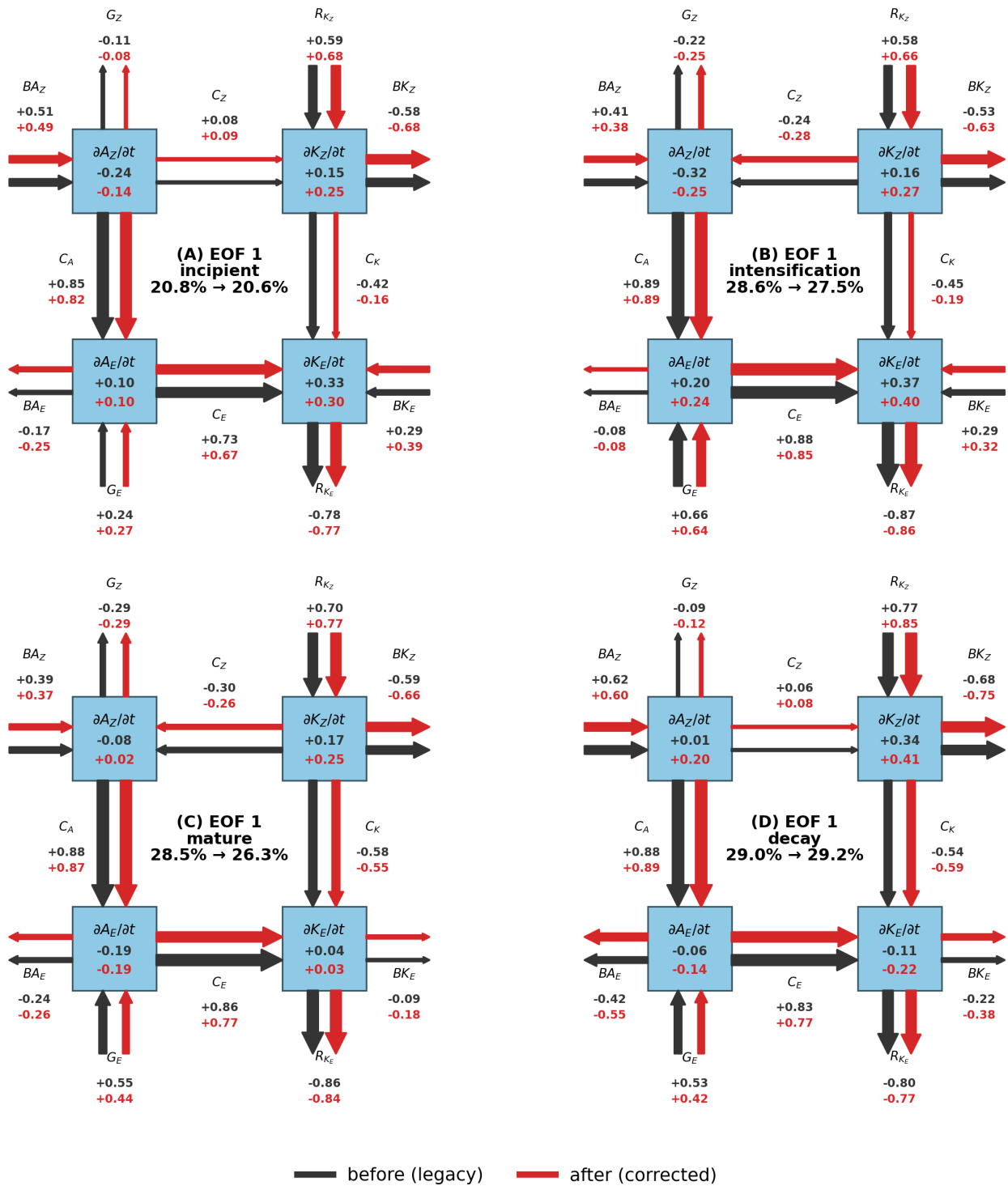
Phase-mean Lorenz Energy Cycle before and after mean $\pm$ sample standard deviation



Primary-phase mean LEC. Dark arrows and values show the published legacy population; red shows the corrected population.

Figure 5

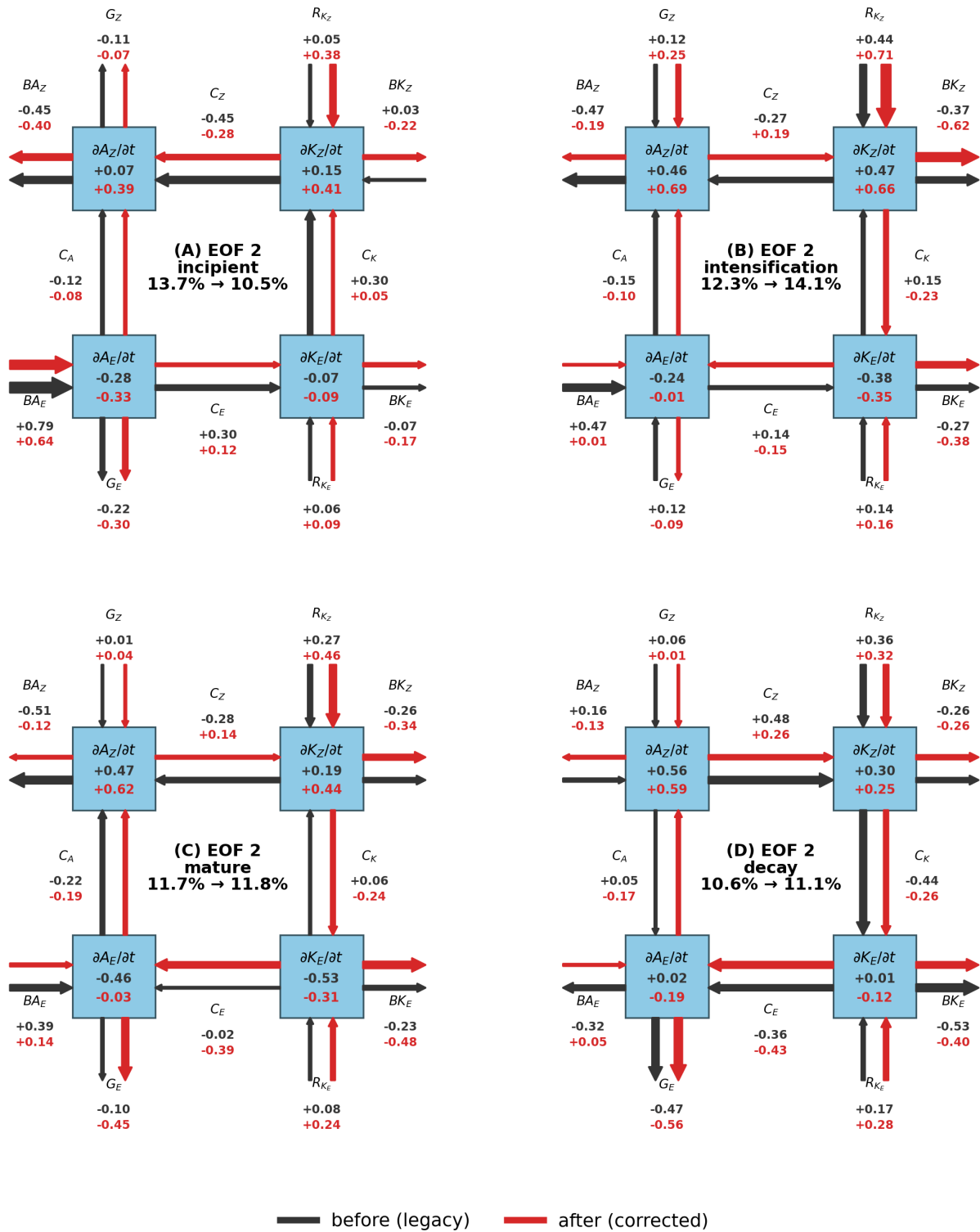
EOF 1 loadings before and after



EOF 1 LEC loadings fitted independently to the published and corrected populations, then pattern-matched and sign-aligned.

Figure 6

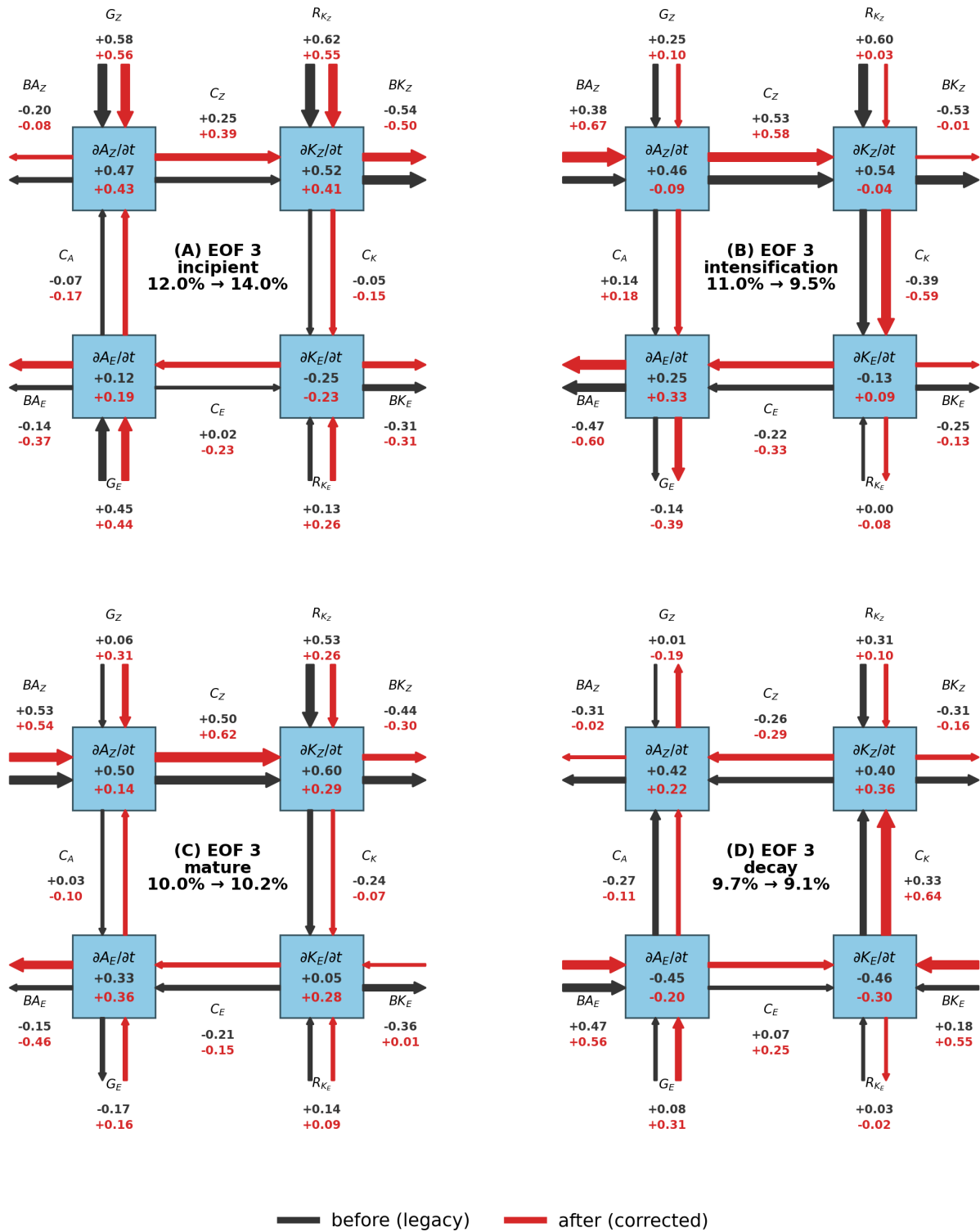
EOF 2 loadings before and after



EOF 2 LEC loadings fitted independently to the published and corrected populations, then pattern-matched and sign-aligned.

Figure 7

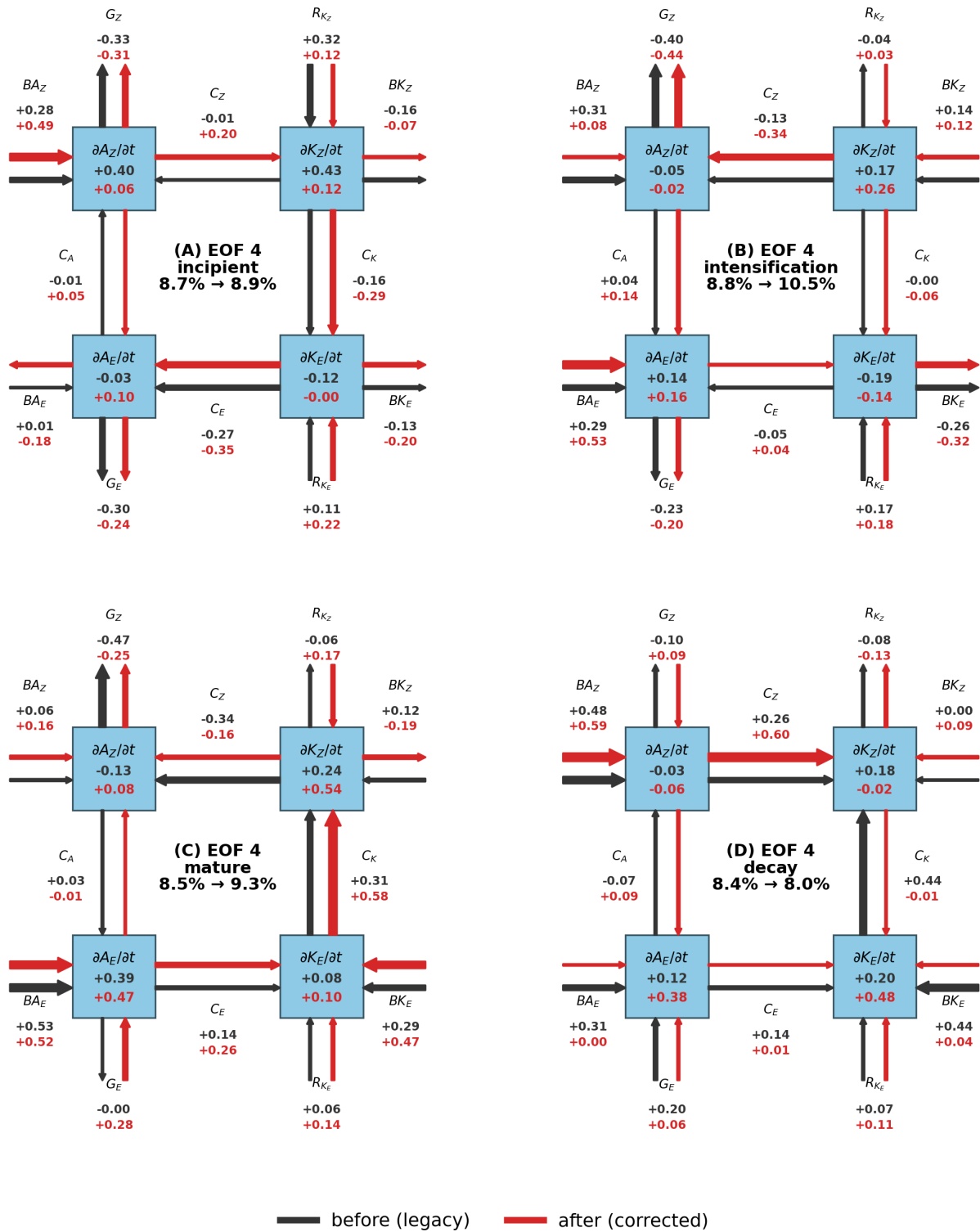
EOF 3 loadings before and after



EOF 3 LEC loadings fitted independently to the published and corrected populations, then pattern-matched and sign-aligned.

Figure 8

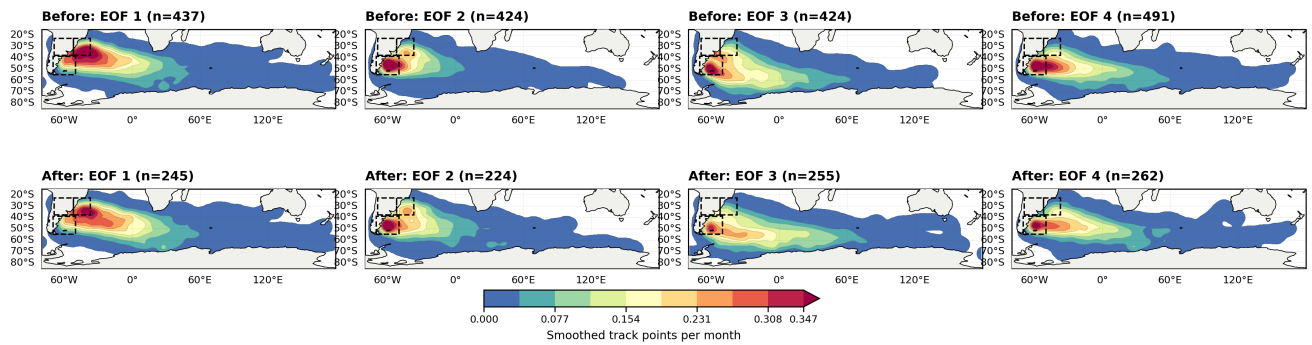
EOF 4 loadings before and after



EOF 4 LEC loadings fitted independently to the published and corrected populations, then pattern-matched and sign-aligned.

Figure 9

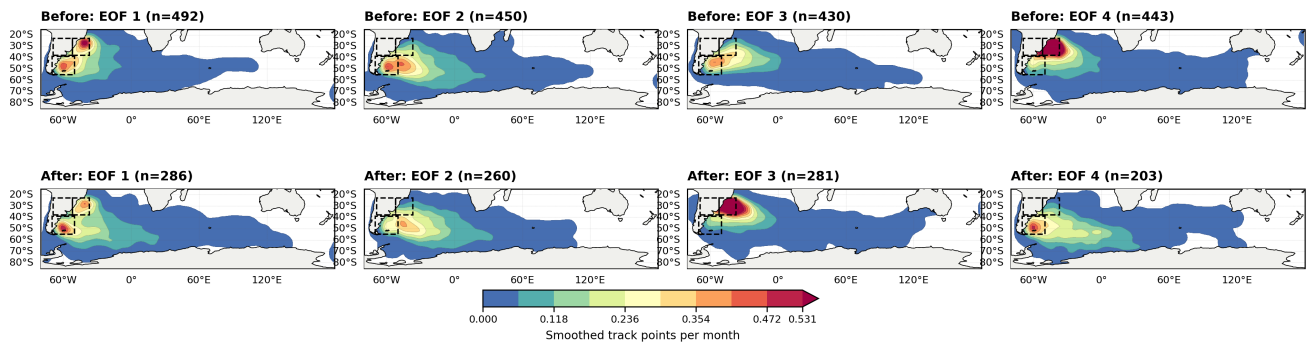
Track density - positive total-lifecycle PC extremes



Positive total-lifecycle PC-extreme track densities: published population above and corrected population below.

Figure 10

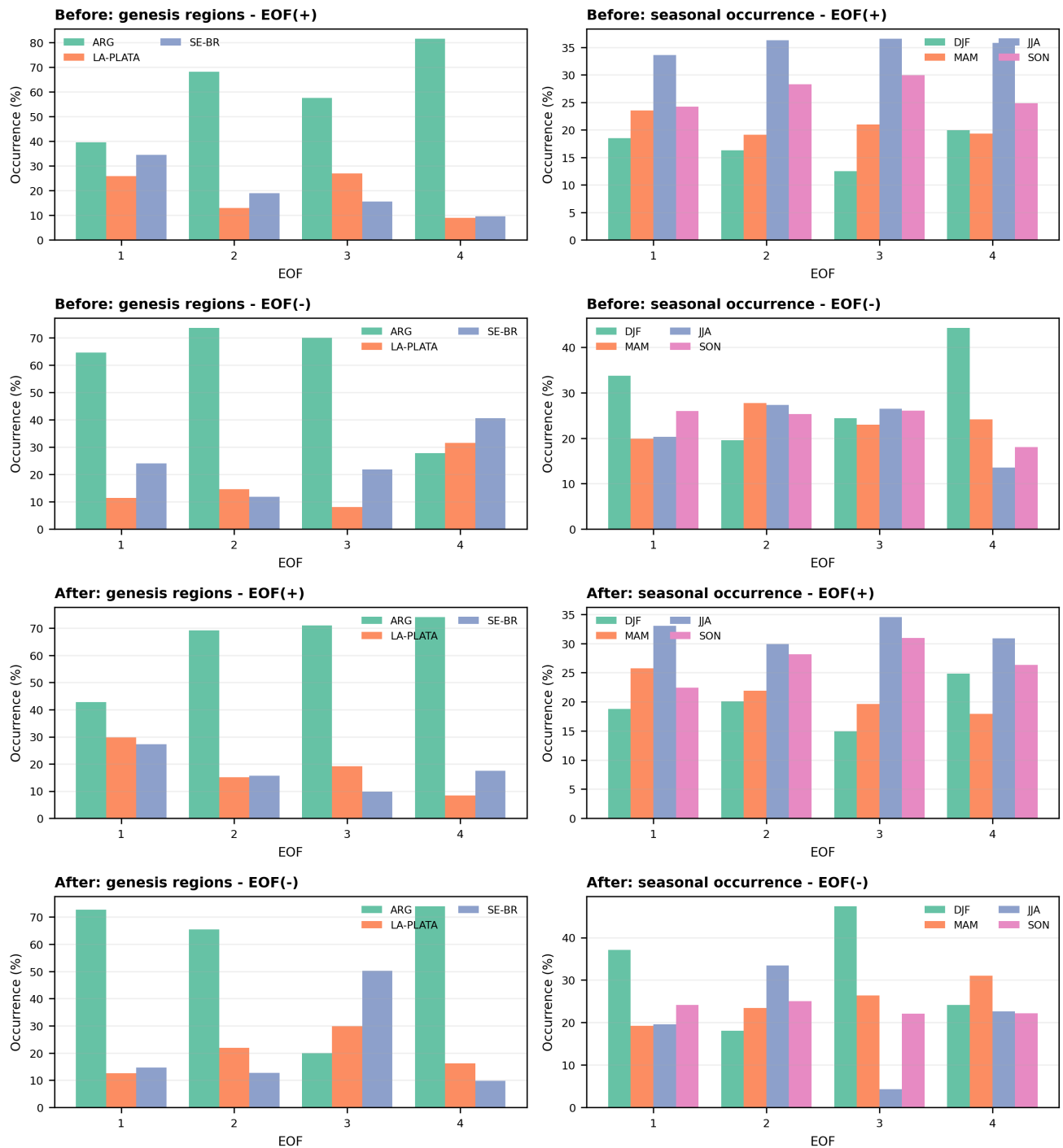
Track density - negative total-lifecycle PC extremes



Negative total-lifecycle PC-extreme track densities: published population above and corrected population below.

Figure 11

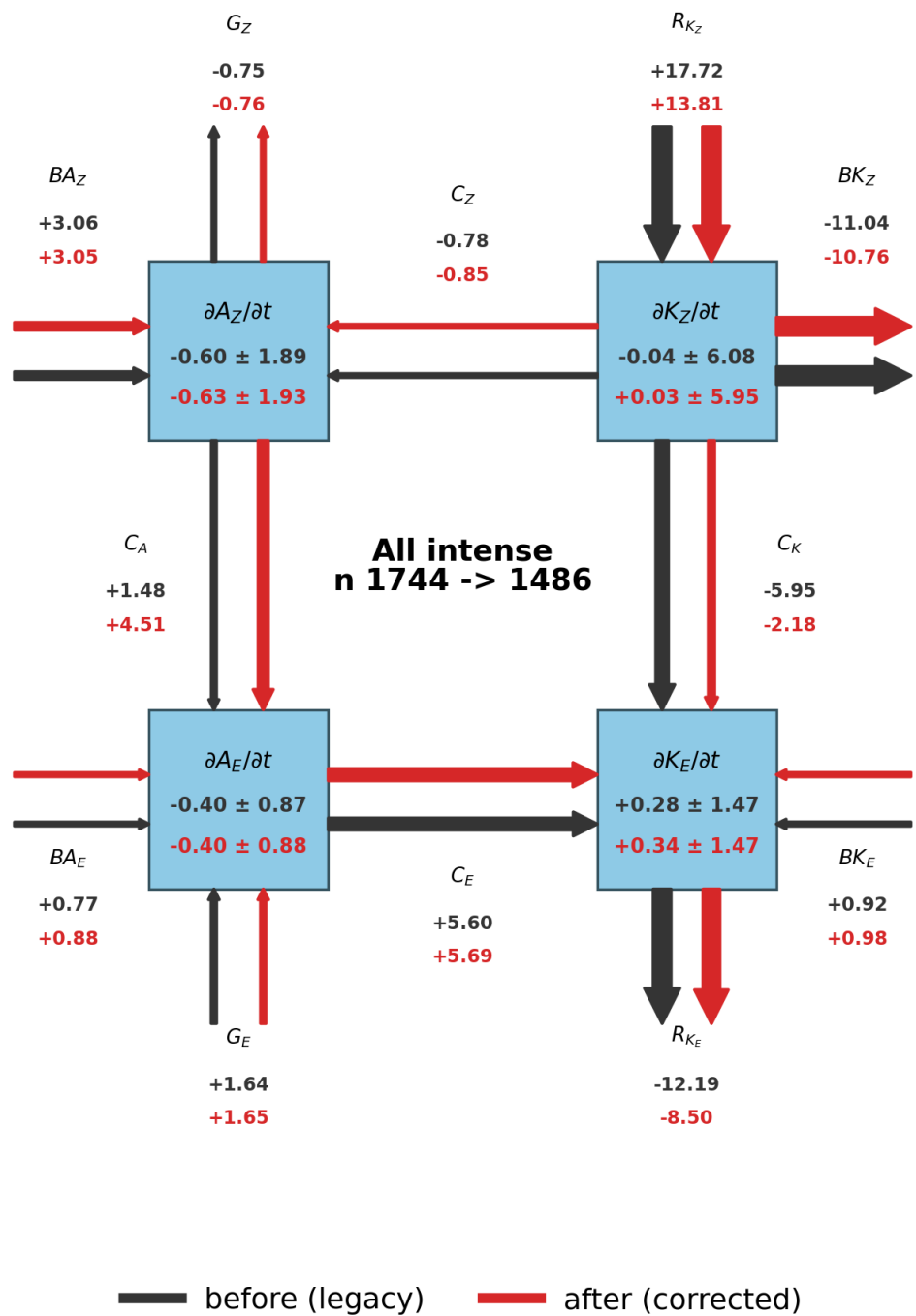
Genesis and seasonal composition of total-lifecycle EOF extremes



Genesis-region and seasonal composition of total-lifecycle PC extremes, published above and corrected below.

Figure 12a

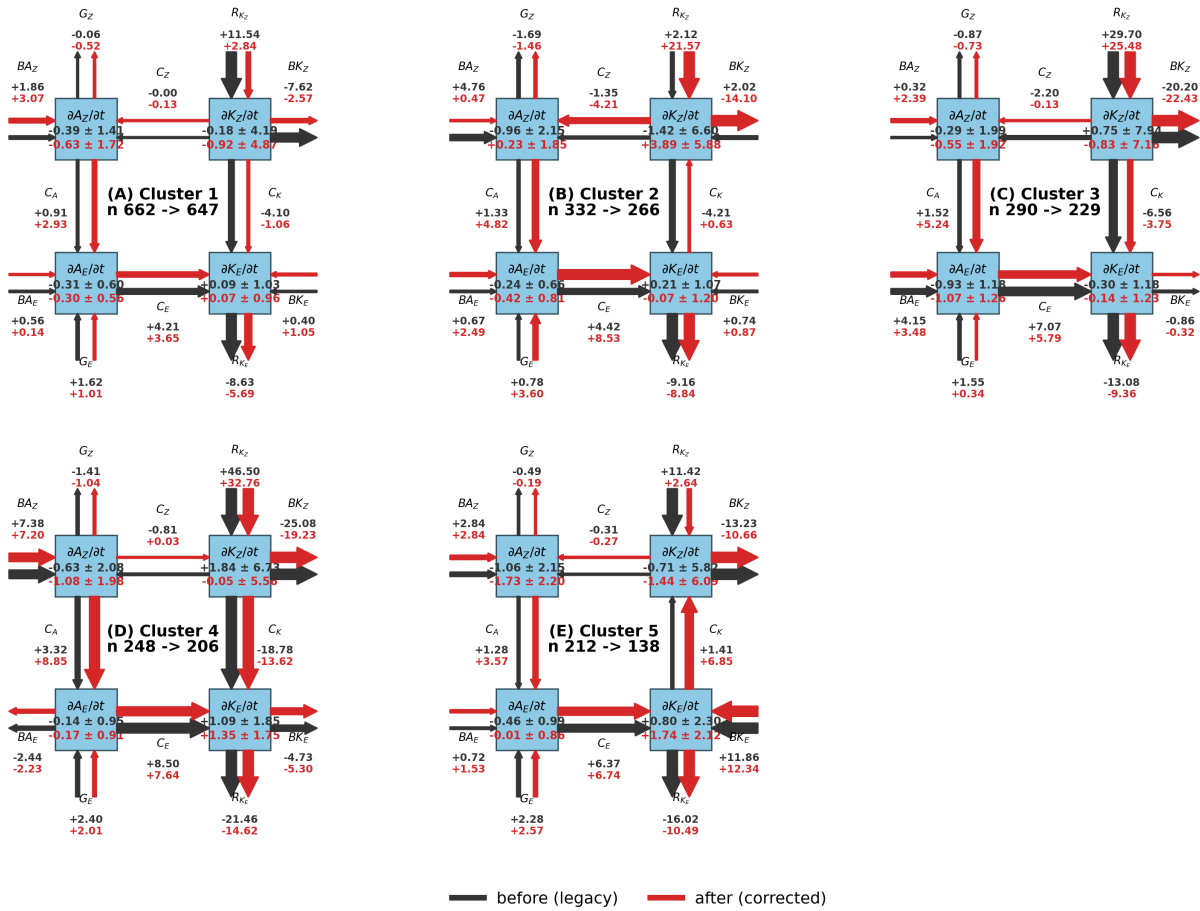
All intense cyclones: published versus corrected



Mean LEC of all intense cyclones in the published and corrected populations.

Figure 12b

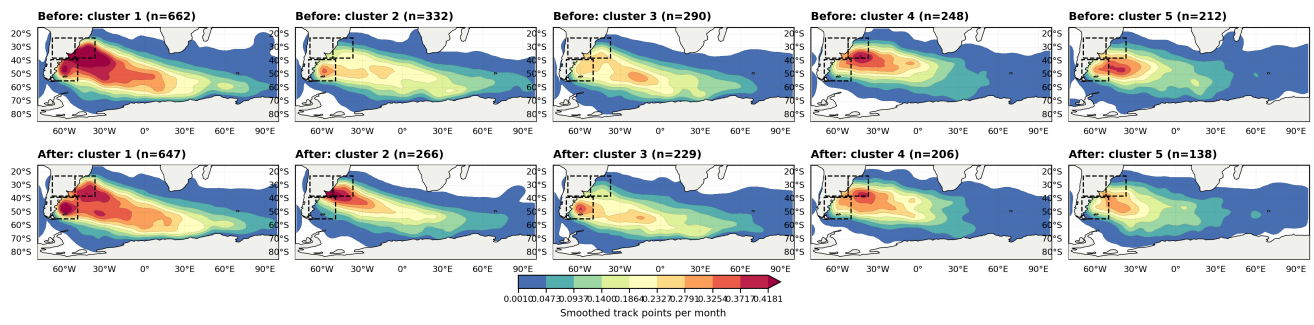
Five intense-cyclone PC groups: published versus corrected



Five intense-cyclone groups obtained from the first eight total-lifecycle PCs; corrected groups are centroid-matched to published groups.

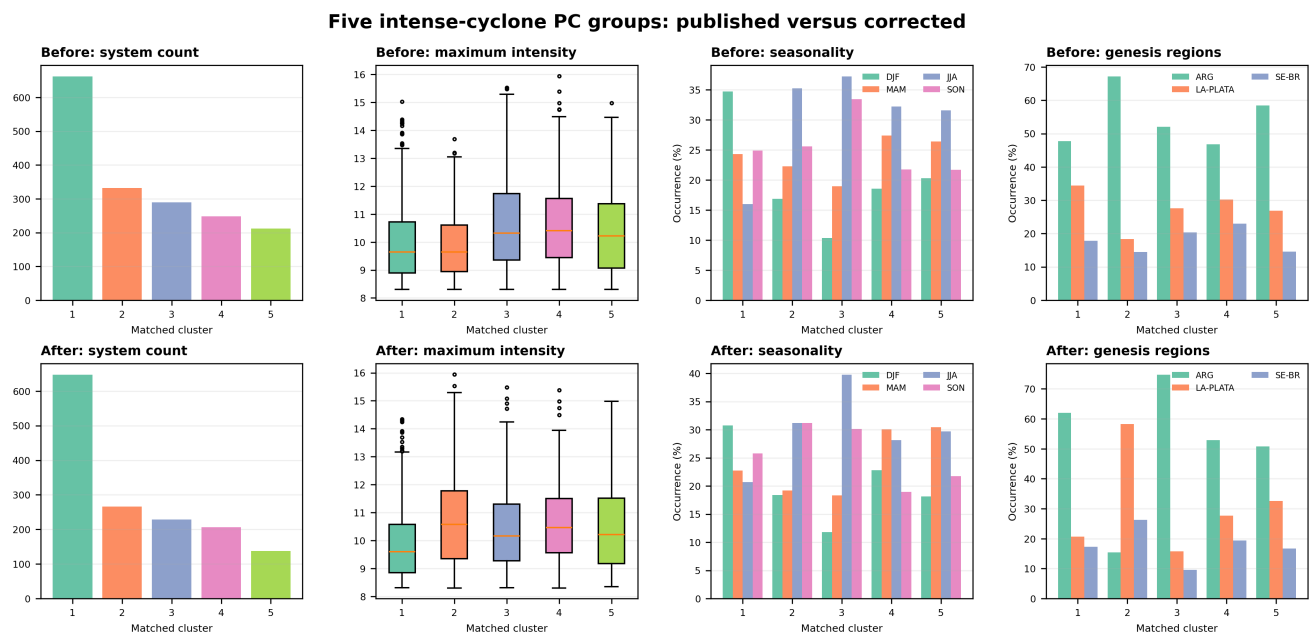
Figure 13

Track density of five intense-cyclone PC groups



Track densities of the five intense-cyclone PC groups, published above and corrected below.

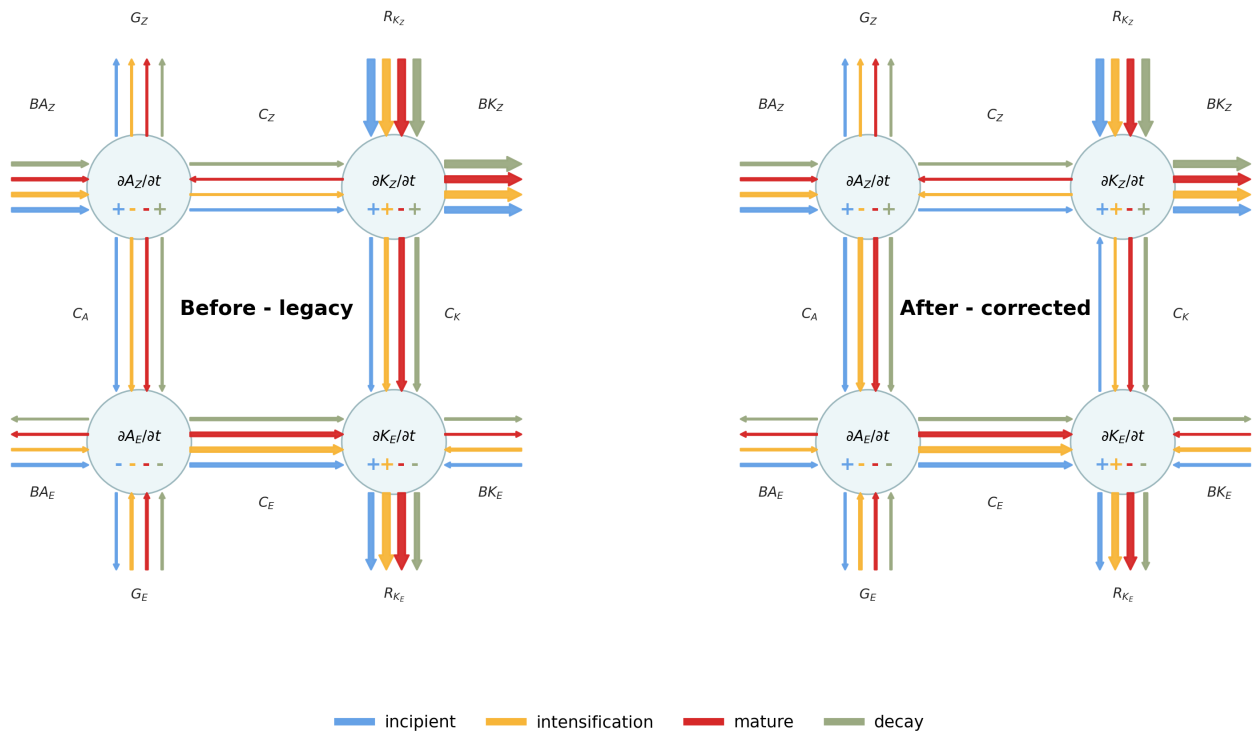
Figure 14



Counts, maximum intensity, seasonality and genesis regions for the five intense-cyclone PC groups.

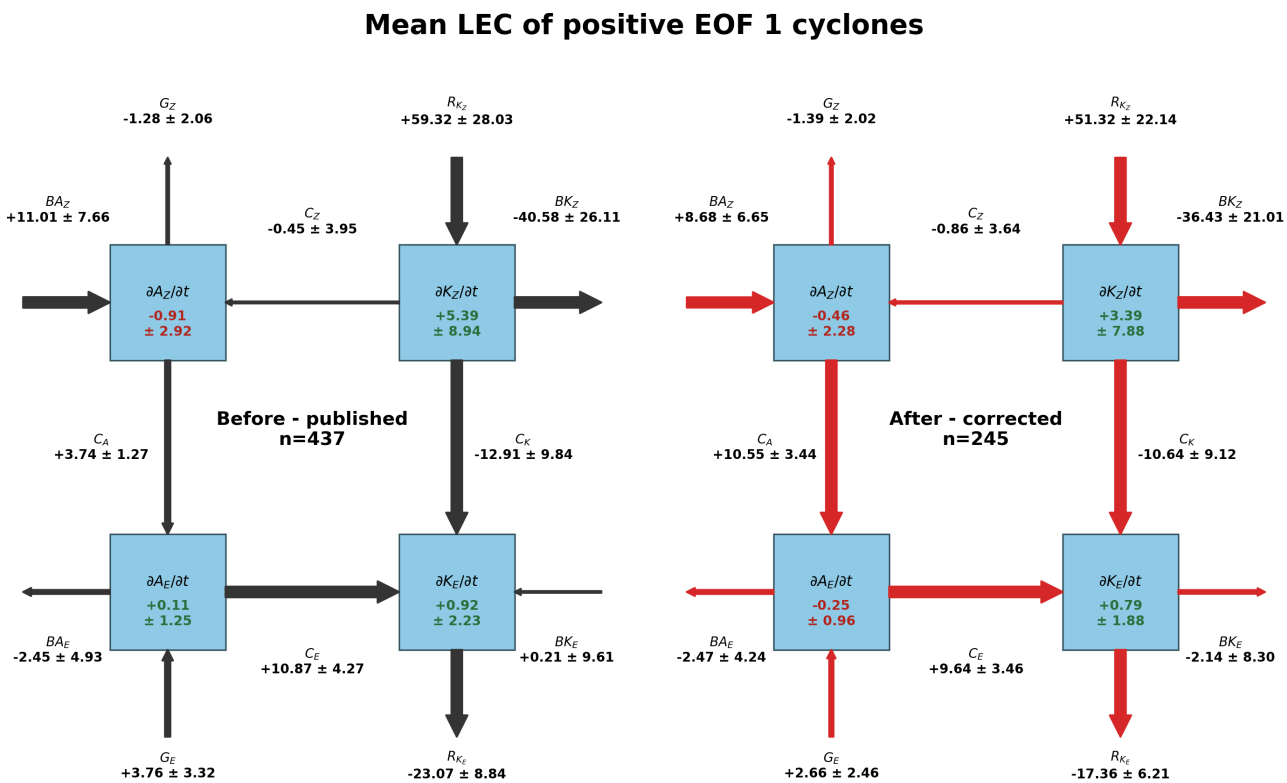
Figure 15

Phase-mean LEC synthesis: before versus after



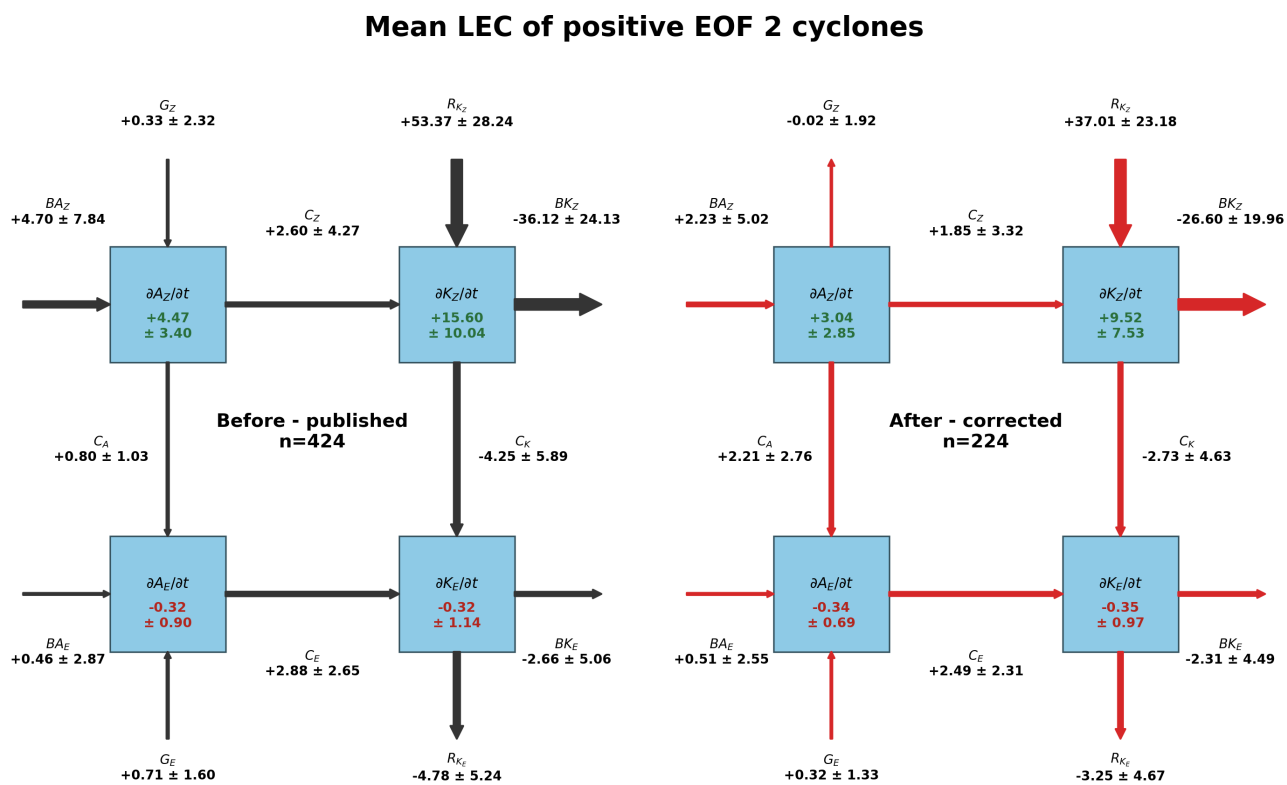
Primary-phase LEC synthesis, with the published population on the left and corrected population on the right.

Figure 16a



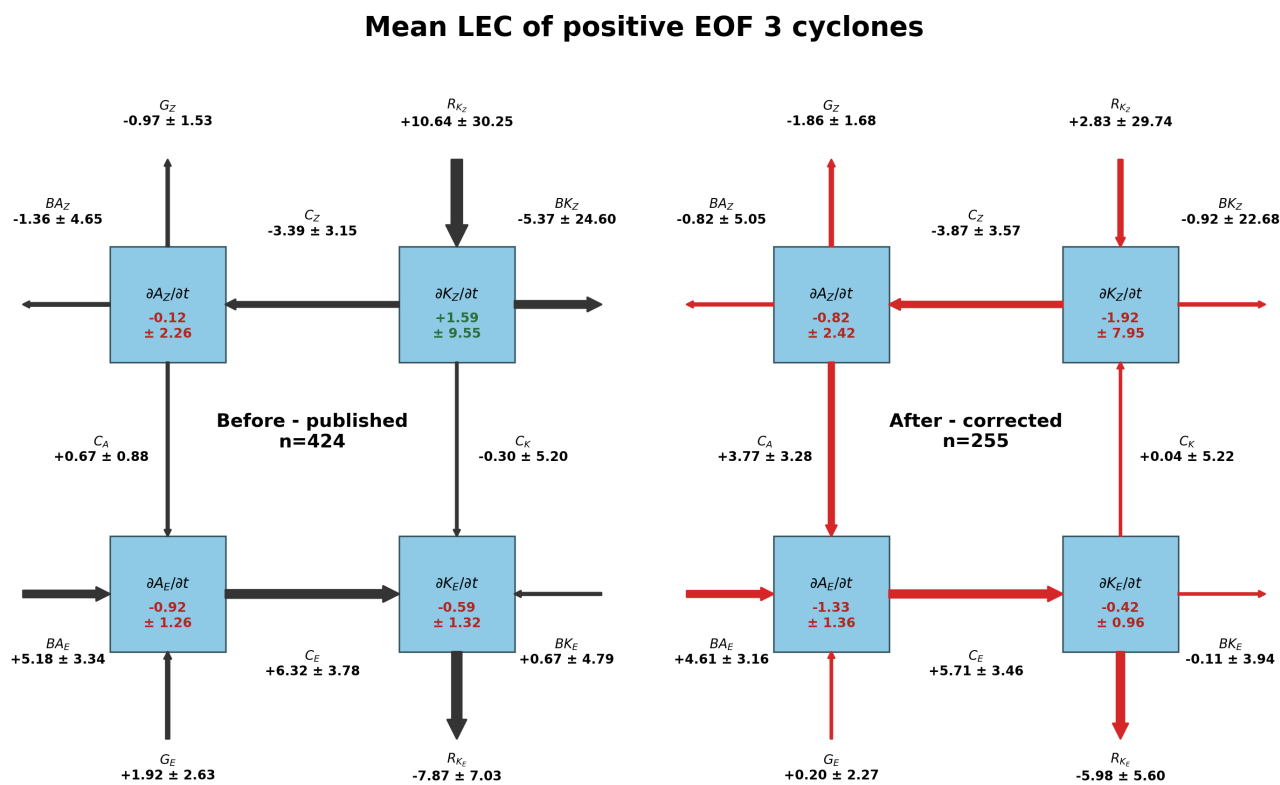
Mean LEC of cyclones assigned to the positive EOF 1 extreme, published on the left and corrected on the right.

Figure 16b



Mean LEC of cyclones assigned to the positive EOF 2 extreme, published on the left and corrected on the right.

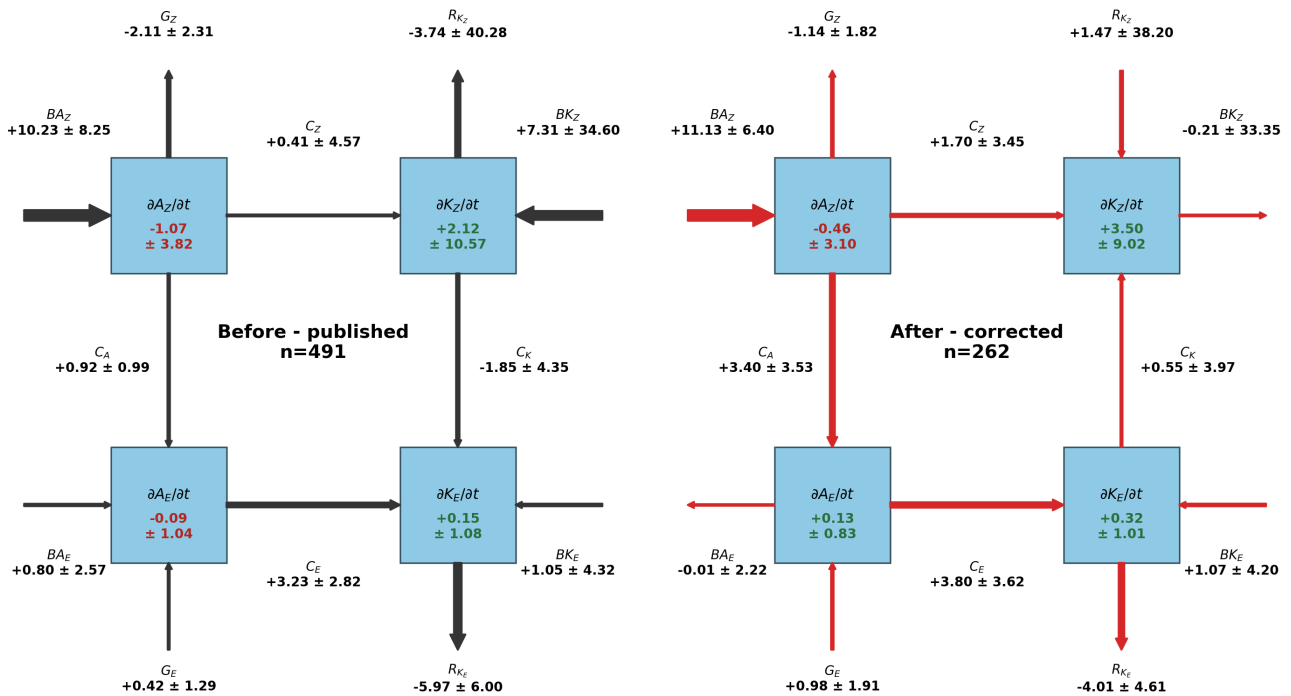
Figure 16c



Mean LEC of cyclones assigned to the positive EOF 3 extreme, published on the left and corrected on the right.

Figure 16d

Mean LEC of positive EOF 4 cyclones



Mean LEC of cyclones assigned to the positive EOF 4 extreme, published on the left and corrected on the right.

Reproducibility record

Input and output SHA-256 hashes, population counts, EOF matching diagnostics, cluster centers, assignments and the audited source commit are recorded under results/article_comparison. The paired control is kept separately under docs/paired_control/.